

ME 514: Fundamentals of Multiphase Flow

Course Code	ME 514
Course Name	Fundamentals of Multiphase Flow
L-T-P-C	3-0-0-3
Prerequisites	ME 210 and ME 303 or equivalent courses
Intended for	M.Tech/MS/PhD/UG
Mutual Exclusion	None
Approval	recommended by the 39th BoA (Mar 2021)

Course Contents

- **Introduction and flow regimes:** Introduction to multiphase flows, equations of motion, flow regimes in horizontal and vertical pipes, flow regimes with phase change. [6 Lectures]
- **Modelling techniques for multiphase flow:** Homogeneous flow models, drift flux models, separated flow models, dispersed phase modelling, modelling of annular and stratified flow. [12 Lectures]
- **Computational techniques:** Introduction to computational modelling of multiphase flow, population balance modelling for dispersed phase, interface tracking methods for multiphase flow (Volume of Fluid and Level set), particle methods for multiphase flow (Lagrangian Point Particle and Smoothed Particle Hydrodynamics), molecular dynamics. [10 Lectures]
- **Experimental techniques:** Introduction to experimental techniques, importance of measurement and experimentation, calibration, uncertainty analysis, error propagation, pressure, velocity and temperature measurements, measurement of size distribution for dispersed phase, measurement of void-fraction and interface reconstruction, data acquisition and analysis techniques. [7 Lectures]
- **Special topics in multiphase flow:** Granular flows, slurry transport, lattice Boltzmann method, phase change heat transfer, micro-fluidics, multiphase flow in micro-channels. [7 Lectures]

Text Books

1. Christopher E. Brennen, **Fundamentals of Multiphase Flow**, Cambridge University Press, 2005.
2. S. Mostafa Ghiaasiaan, **Two-Phase Flow, Boiling, and Condensation**, 2nd Edition, Cambridge University Press, 2007.

References

1. John G. Collier and John R. Thome, **Convective Boiling and Condensation**, Oxford University Press; 3rd Edition, 1994.
2. L. S. Tong and Y. S. Tang, **Boiling Heat Transfer and Two-Phase Flow**, 2nd Edition, CRC Press, 2010.
3. R. Clift, J. R. Grace and M. E. Weber, **Bubbles, Drops, and Particles**, Dover Publications, Inc., 2005.
4. Mamoru Ishii and Takashi Hibiki, **Thermo-Fluid Dynamics of Two-Phase Flow**, 2nd Edition, Springer, 2011.
5. J. P. Holman, **Experimental Methods for Engineers**, 7th Edition, Tata McGraw-Hill, 2001.
6. Andrea Prosperetti and Grtvar Tryggvason, **Computational Methods for Multiphase Flow**, Cambridge University Press, 2007.
7. Greta Tryggvason, Ruben Scardovelli and Stephane Zaleski, **Direct Numerical Simulations of Gas-Liquid Multiphase Flows**, Cambridge University Press, 2011.